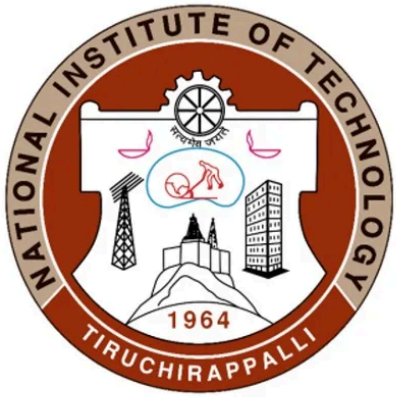



NATIONAL INSTITUTE OF
TECHNOLOGY, Tiruchirappalli

# CSPE 72

## Deep Learning Techniques

# Automatic Question Generation from Study Material

PROJECT REPORT

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# Automatic Question Generation from Study Material


## Abstract

Preparing practice questions from lecture material is a time-consuming task for instructors and learners. This report presents StudyQG, a system that generates verified, examination-style questions from study material supplied as a PDF file or as plain text. The system segments the material into passages, selects candidate answer spans with a combination of linguistic heuristics and a neural answer extractor, generates a question for each span with a T5 sequence-to-sequence Transformer fine-tuned on SQuAD v1.1, verifies every question through a round-trip check with an extractive question-answering model, and formats the verified questions as short-answer, multiple-choice, cloze and true/false items. A T5-small model fine-tuned by the team on 8,000 SQuAD examples in 45 minutes on a CPU attains BLEU-4 17.0, ROUGE-L 42.8 and METEOR 41.5 on a test set of 300 held-out paragraphs, against 2.7, 15.2 and 10.2 before fine-tuning; the publicly available T5-base checkpoint used in the demonstration attains 24.5, 49.7 and 49.5 under the same protocol. The complete system runs offline on a laptop without a GPU.


## 1. Team Details


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Course: CSPE72 Deep Learning Techniques. Assigned problem statement: Automatic Question Generation from Study Material (Team 3).

## 2. Problem Statement

Instructors and learners spend considerable time composing practice questions by hand. The task addressed in this project is the automatic generation of examination-style questions from study material supplied in a common format: lecture notes as a PDF file, a textbook chapter as text, or pasted paragraphs. The system is required to produce well-

formed, answerable questions that cover the material in four formats (short answer, multiple choice with plausible distractors, fill in the blank, and true/false), each accompanied by its answer and by the source sentence from which it was derived.

Formally, given a document D, the system produces a set of triples (q, a, t) such that the answer a is a span of D, the question q is answerable from D, the item type t is one of the four formats, the set covers the principal facts of D, and near-duplicate questions are removed. The central learning problem is answer-aware question generation: modelling p(q | p, a) for a passage p and an answer span a with a sequence-to-sequence Transformer fine-tuned on the SQuAD dataset [1].

## 3. Gaps Identified

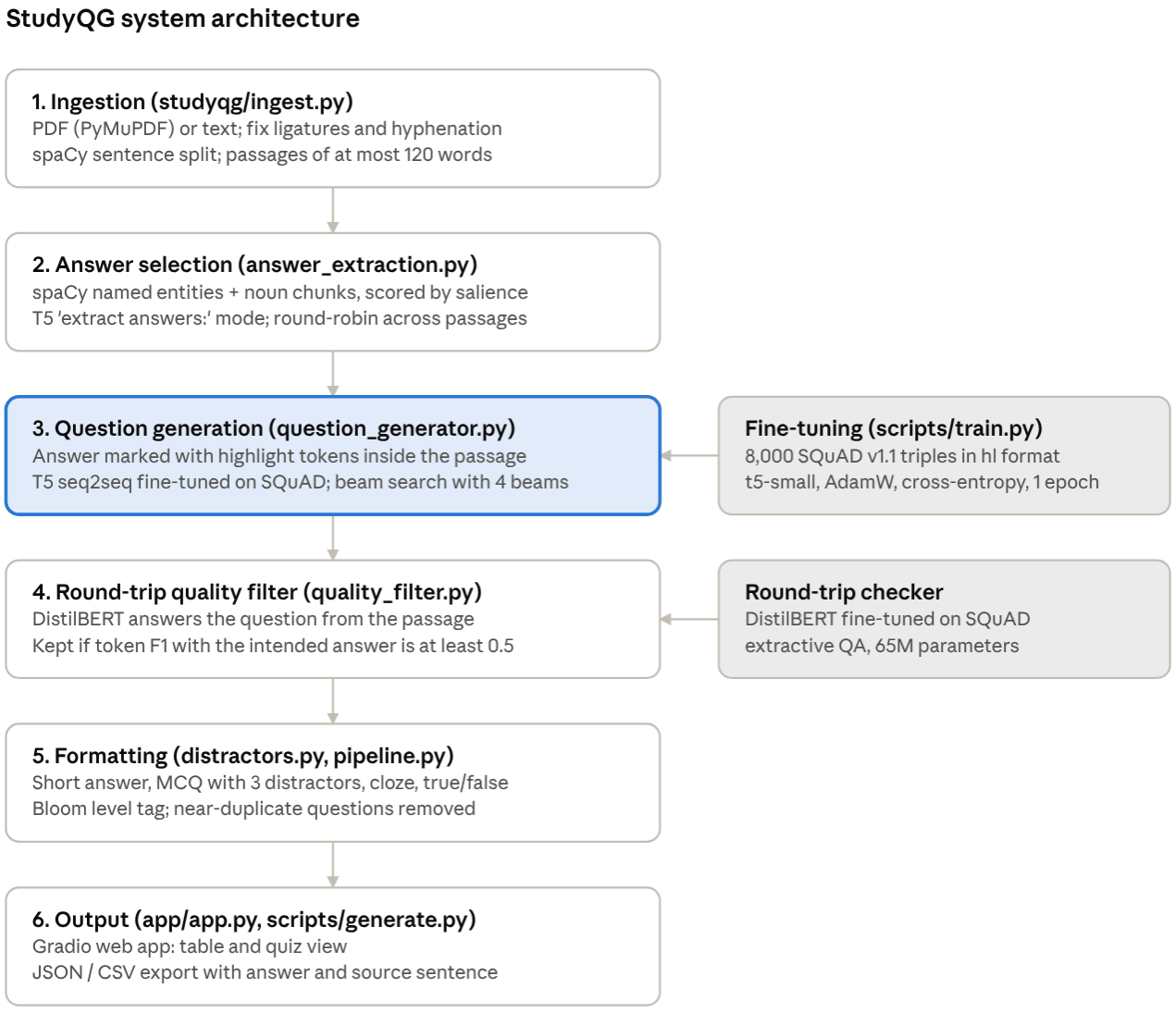
A review of existing approaches reveals five gaps between published question-generation research and a usable tool for study material.

1. Rule-based systems. Early systems transform declarative sentences into questions with hand-written syntactic rules and rank the output statistically [2]. The resulting questions are often stilted, and the rules fail on the long, complex sentences that are typical of lecture notes.
2. Answer-aware models assume that the answer is given. Neural generators trained on SQuAD, from the attention-based LSTM of Du et al. [3] to pre-trained Transformers such as UniLM [4], ProphetNet [5] and the T5 models of Ushio et al. [6], generate a question for a provided answer span. Study material carries no marked answers, so a complete system requires answer selection and a strategy for covering the whole document.
3. Absence of an answerability check. Sequence-to-sequence generators can produce fluent questions that are unanswerable from the passage or that disclose the answer. Published pipelines rarely verify generated questions automatically; round-trip consistency has been used to filter synthetic training corpora [7] but not as an interactive quality gate for learners.
4. A single question format. Most systems produce only wh-questions, whereas assessment practice also requires multiple-choice items with plausible distractors, cloze items and true/false statements.
5. The clean-text assumption. Research code expects clean paragraphs, while real material arrives as PDF files with running headers, page numbers, hyphenated line breaks and typographic ligatures.

The system described in this report addresses each of these gaps: a T5 generator fine-tuned on SQuAD, a hybrid answer selector with document-level coverage, a round-trip verification stage, four item formats, and a PDF ingestion stage.

# 4. Proposed Architecture

The system is organised as a six-stage pipeline (Figure 1) built around two learned components: a T5 question generator fine-tuned on SQuAD, and a DistilBERT extractive question-answering model used for verification. The remaining stages are deterministic and serve to prepare the input for the generator and to package its output.




StudyQG architecture: 6 inference stages, 2 trained models

Figure 1. System architecture of StudyQG. The six stages on the left are executed for every input document; the two components on the right are the learned models.

Stage 1 (ingestion, module ingest.py) extracts text from PDF files with PyMuPDF, normalises ligatures and quotation marks, repairs hyphenated line breaks, discards lines that are predominantly non-alphabetic, splits the text into sentences with spaCy and groups sentences into passages of at most 120 words without crossing paragraph boundaries. Stage 2 (answer selection, answer_extraction.py) proposes candidate answer spans from named entities and noun chunks, scores them by entity type, syntactic role

and document frequency, adds spans proposed by the T5 model in its answer-extraction mode, and selects candidates round-robin across passages so that the generated questions cover the whole document. Stage 3 (question generation, question_generator.py) marks the selected span with highlight tokens inside its passage and generates a question with beam search. Stage 4 (verification, quality_filter.py) submits each generated question together with its passage to the DistilBERT model and retains the question only if the token-level F1 between the predicted span and the intended answer is at least 0.5. Stage 5 (formatting, distractors.py and pipeline.py) produces the four item types, generates distractors, removes near-duplicates and assigns a Bloom taxonomy level. Stage 6 (output, app/app.py and scripts/generate.py) presents the items in a web interface and exports them as JSON and CSV. The question generator of stage 3 is the deep learning core of the system.

## 5. Algorithmic Steps

Algorithm 1. Fine-tuning of the question generator (scripts/train.py).

1. Load SQuAD v1.1 [1] and convert every (context, answer, question) triple into highlight format: the answer span is enclosed in `<hl>` tokens within the context and the task prefix "generate question:" is prepended. Sample 8,000 training and 300 validation examples with a fixed random seed.
2. Load the pre-trained t5-small model [8], register `<hl>` as an additional special token and resize the embedding matrix accordingly.
3. For each mini-batch of 8 examples, tokenise the source (at most 384 tokens) and the target question (at most 48 tokens); encode the source and decode the right-shifted target with teacher forcing; compute the token-level cross-entropy loss against the target.
4. Back-propagate the loss, clip the gradient norm to 1.0 and update the parameters with AdamW (learning rate 3e-4, weight decay 0.01) under a schedule of 5% linear warm-up followed by linear decay.
5. Every 100 optimiser steps, evaluate the cross-entropy loss on the validation set (200 held-out training-split examples) and on the test set (300 development paragraphs); at the end of training save the model and tokenizer.

Algorithm 2. Question generation for a document (studyqg/pipeline.py).

1. Ingestion: read the PDF or text, normalise ligatures and quotation marks, repair hyphenated line breaks, discard predominantly non-alphabetic lines (page numbers, rules), split the text into sentences with spaCy, and group sentences into passages of at most 120 words without splitting a sentence or a paragraph.

2. Candidate answers: for every sentence, collect named entities (persons, dates, organisations, quantities) and noun chunks with spaCy; score each candidate by type and syntactic role and add a salience bonus proportional to the logarithm of its frequency in the document. When the lmqg checkpoint is used, also apply T5 in "extract answers:" mode to each sentence. Remove duplicates within a passage.
3. Selection: choose candidates round-robin across passages, one per sentence first, until 1.8 N candidates have been selected for a request of N questions, so that the questions cover the whole document.
4. Generation: highlight each selected answer within its passage and apply T5 with beam search (4 beams, no repeated trigrams, at most 64 new tokens). Normalise the output (capitalisation, terminal question mark), discard near-duplicates (string similarity above 0.9) and discard questions that contain the answer verbatim.
5. Verification: submit each question with its passage to the DistilBERT-SQuAD model; compute the token-level F1 between the predicted span and the intended answer; retain questions with F1 of at least 0.5, rank them by (F1, model confidence, candidate score), keep the top N and restore document order.
6. Formatting: cycle through the requested item types. Multiple choice: three distractors drawn from same-type spans of the document, WordNet co-hyponyms or perturbed numbers, shuffled with the answer. Fill in the blank: the source sentence with the answer removed. True/false: the source sentence unchanged, or with the answer replaced by a distractor. Assign a Bloom taxonomy level from the wording of the question.
7. Output: tabular view, quiz view, and JSON and CSV files containing the answer, quality score and source sentence of every item.

## 6. Dataset Used

| Dataset | Use | Size | Source |
|---|---|---|---|
| SQuAD v1.1 [1] | Fine-tuning and evaluation of the question generator | 87,599 training and 10,570 validation triples from 536 Wikipedia articles; 8,000 training examples and 200 disjoint validation examples were sampled from the training split, and 300 paragraphs (one question each) from the development split form the test set | rajpurkar/squad on the Hugging Face Hub |
| Sample study material | Demonstration inputs | Three passages of approximately 350 words written by the team on photosynthesis, neural networks and the Indian space programme | data/sample/ in the repository |

Table 1. Datasets used in the project.

SQuAD provides (context paragraph, question, answer span) triples. The task is reversed in this project: the model receives the context with the answer span highlighted and is trained to produce the question. The same dataset underlies the public lmqg checkpoints [6] and the DistilBERT question-answering model [9] used for verification, so every learned component shares one annotation convention.

## 7. Tools Used

| Tool | Version | Purpose |
|---|---|---|
| Python | 3.12 | Implementation language |
| PyTorch (CPU build) | 2.14 | Model training and inference |
| Hugging Face Transformers [12] and Datasets | 5.21, 5.1 | T5 and DistilBERT models, tokenizers, beam-search decoding; loading SQuAD v1.1 |

| Tool | Version | Purpose |
|---|---|---|
| spaCy with en_core_web_sm [11] | 3.8 | Sentence segmentation, named entities, noun chunks |
| NLTK, sacrebleu, rouge-score | 3.10, 2.6 | METEOR, BLEU-4 and ROUGE-L metrics; WordNet distractors |
| PyMuPDF | 1.27 | PDF text extraction |
| Gradio | 6.31 | Web interface |
| pandas, matplotlib | 3.1, 3.11 | Result tables and plots |
| Git and GitHub | | Version control and code submission |
| Claude Code (Anthropic) | | AI coding assistant; see Section 14 |

Table 2. Software and hardware.

All experiments, including fine-tuning, were executed on a laptop with a 16-core CPU, 15 GB of RAM and no GPU (Ubuntu under WSL2). PyTorch was restricted to 8 threads, which was measured to be substantially faster than 16 threads on this hardware.

## 8. Screenshots of the Result

Figures 2 and 3 show the web interface after generating ten questions from the neural-networks sample (398 words, four passages). Sixty-six candidate answers were extracted, eighteen questions were generated and all eighteen passed the round-trip verification; the mean quality score was 1.0. The reported wall-clock time of 157.6 s was measured while a training job shared the CPU. Figures 4 to 6 show the training and validation loss of the fine-tuned model, its loss on the test set, and the metric comparison discussed in Section 9.

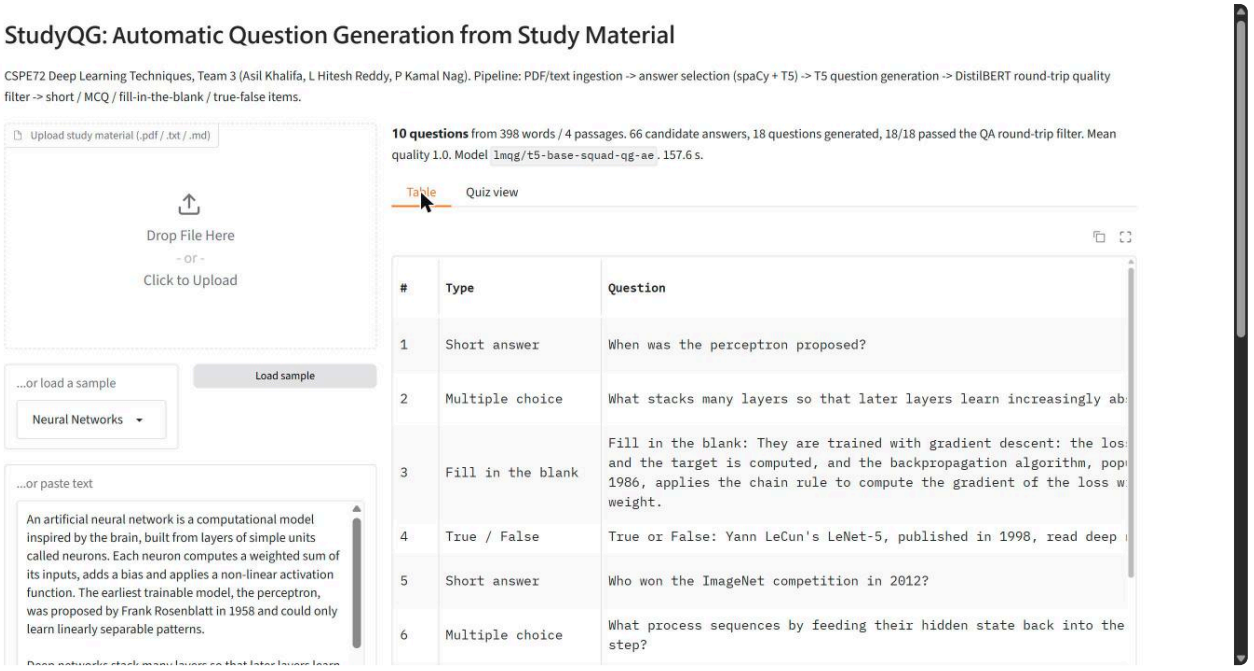



Figure 2. Tabular view of the generated questions in the web interface, with item type, question, options, answer, quality score, Bloom level and source sentence.

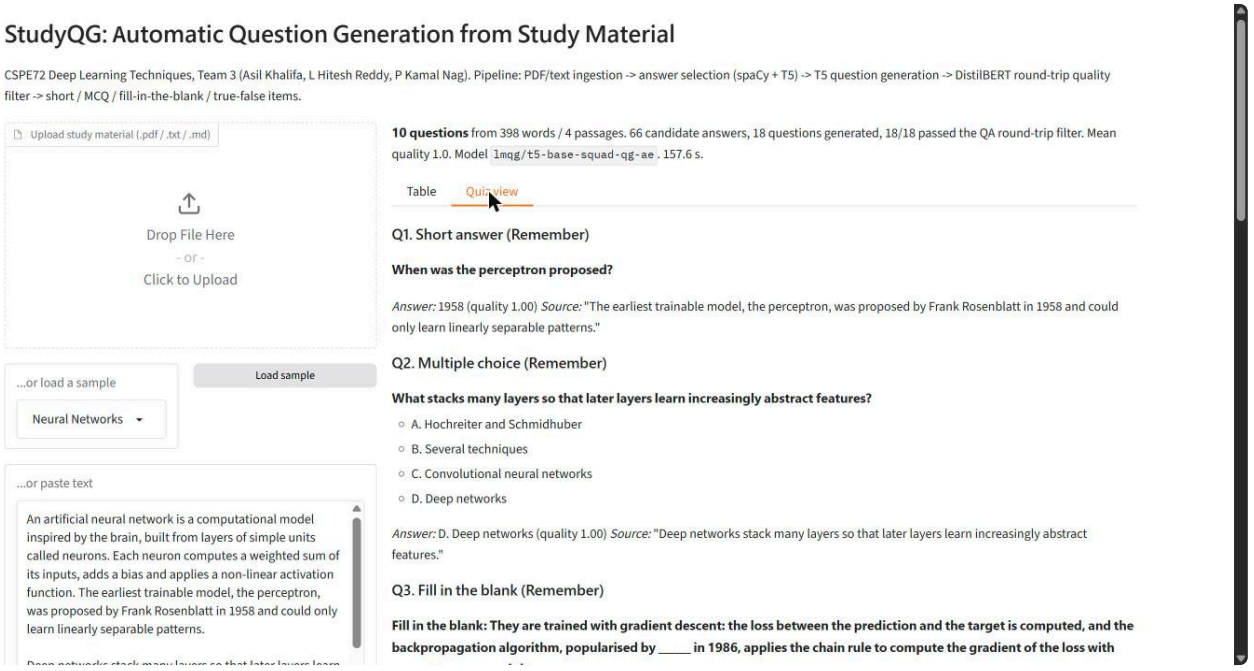



Figure 3. Quiz view with the answer key, the quality score and the source sentence of each item. Item Q4 is a true/false statement constructed by substituting a distractor for the verified answer.

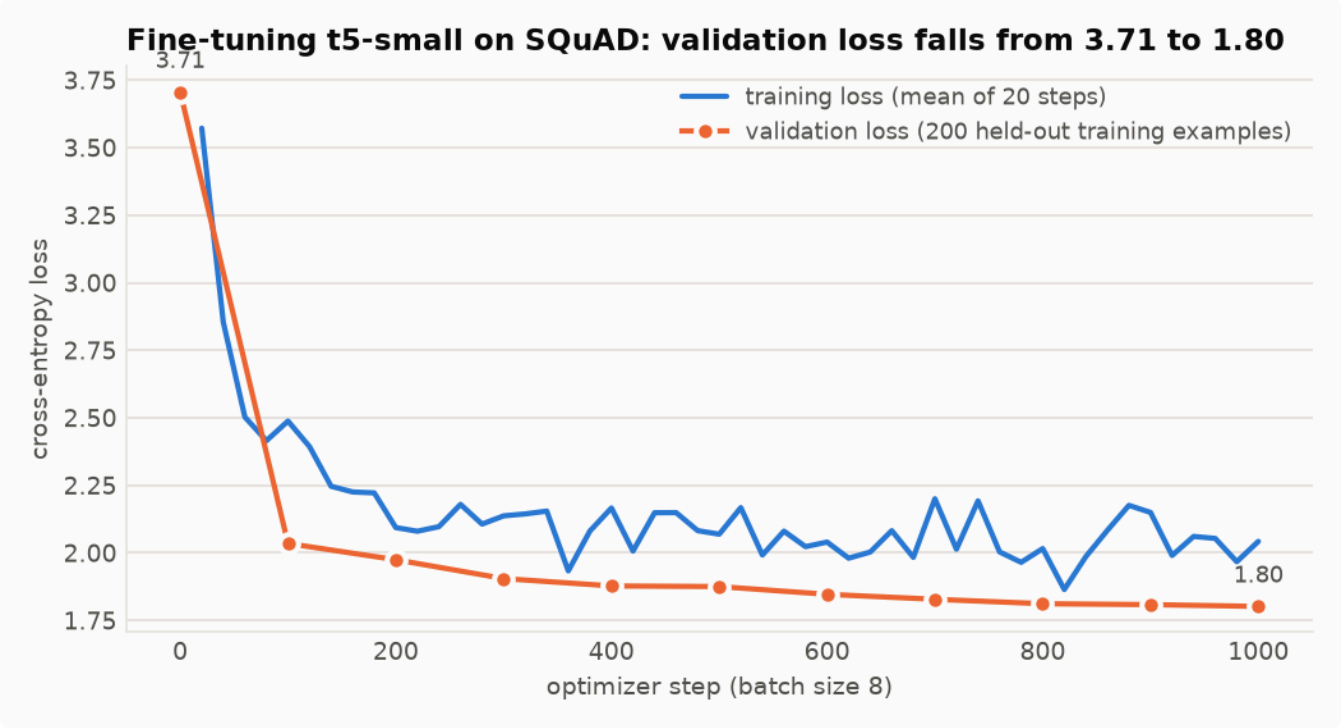



Figure 4. Training loss and validation loss during fine-tuning of t5-small on 8,000 SQuAD examples (1,000 optimiser steps, batch size 8, 45 minutes on the CPU). The validation loss, measured every 100 steps on 200 held-out training-split examples, decreases from 3.71 to 1.80.

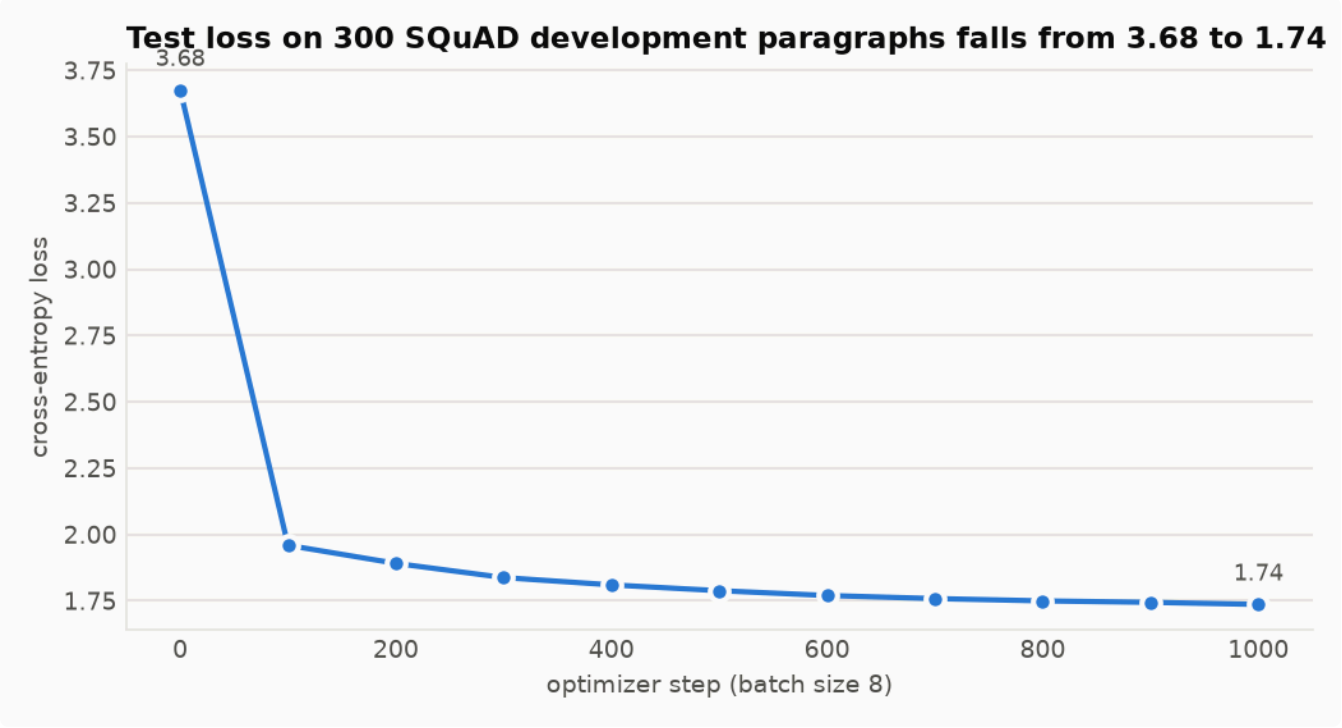



Figure 5. Loss on the test set (300 paragraphs of the SQuAD development split, one question each) measured every 100 optimiser steps during the same fine-tuning run. The test loss decreases from 3.68 to 1.74 and is still decreasing slowly at the end of the epoch, which indicates that a second epoch would help modestly.

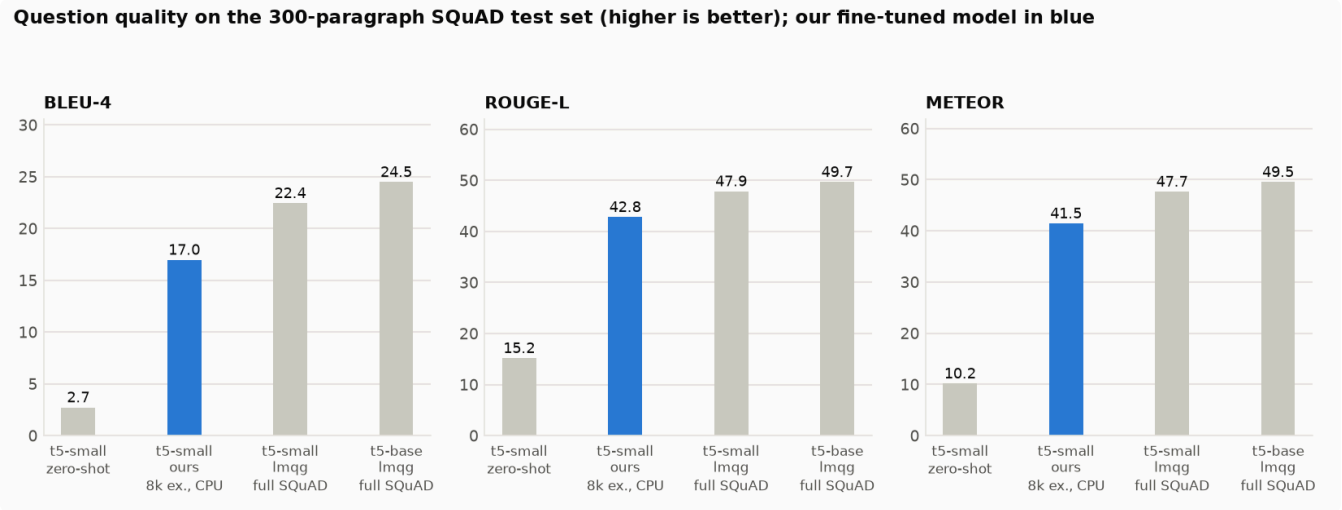



Figure 6. BLEU-4, ROUGE-L and METEOR of the four question generators on 300 SQuAD validation paragraphs. The model fine-tuned by the team is shown in blue.

# 9. Analysis and Comparison with Existing Approaches

## 9.1 Quantitative results

All generators were evaluated on the test set of 300 paragraphs from the SQuAD v1.1 development split (one question per paragraph) with beam size 4, using sacrebleu for BLEU, rouge-score for ROUGE-L and the NLTK implementation of METEOR (scripts/evaluate.py). Scores range from 0 to 100 and higher values are better. Latency is the mean generation time per question on the laptop CPU with 8 threads.

| Model | Training data | BLEU-4 | ROUGE-L | METEOR | Seconds per question |
|---|---|---|---|---|---|
| t5-small, no fine-tuning | none | 2.7 | 15.2 | 10.2 | 0.55 |
| t5-small, fine-tuned in this project | 8,000 SQuAD examples, 1 epoch, 45 min on CPU | 17.0 | 42.8 | 41.5 | 0.38 |
| t5-small, lmqg [6] | full SQuAD (87,599 examples), GPU | 22.5 | 47.9 | 47.7 | 0.91 |
| t5-base, lmqg [6] (demonstration default) | full SQuAD (87,599 examples), GPU | 24.5 | 49.7 | 49.5 | 2.69 |

Table 3. Evaluation of the question generators.

Fine-tuning accounts for most of the attained quality. Without fine-tuning, T5 mostly copies a span of the passage instead of producing a question (for example, "the Teaching Council"). After 1,000 optimiser steps on a CPU, the test loss decreases from 3.68 to 1.74

(validation loss from 3.71 to 1.80) and BLEU-4 increases from 2.7 to 17.0, which closes approximately 80% of the gap to the lmqg t5-small checkpoint trained on eleven times more data with a GPU (72% on BLEU-4, 85% on ROUGE-L and 83% on METEOR).

Training data and model size account for the remainder. Training on the full dataset adds 5.5 BLEU-4 at the same model size, and the move from t5-small to t5-base adds a further 2.0 BLEU-4 and 1.8 ROUGE-L at three times the latency. The demonstration therefore uses t5-base by default and offers t5-small as a faster alternative.

Reference-based metrics underestimate question quality. Exact match with the reference question lies between 1% and 5% even for the strongest model, because a given answer admits many valid questions. For the answer "territory east of the Mississippi", the reference question is "What territory was ceded to Britain?", whereas the fine-tuned model produces "What territory did France cede to Great Britain?", a correct question with a low BLEU score. The round-trip verification stage addresses this limitation without a reference: on the neural-networks sample, 18 of 18 generated questions were answerable by DistilBERT with F1 of at least 0.5; on the ISRO sample, 11 of 15.

Two failure modes were observed. The small model occasionally produces an ungrammatical question (for example, "Who was the seniority of Genghis' oldest son Jochi?") or includes the answer in the question; the pipeline discards questions that contain the answer verbatim and relies on the verification stage for the remaining cases. Distractors for abstract noun phrases are weaker than those for names, dates and numbers.

## 9.2 Comparison with published question-generation models

| System | Year | Architecture | Training data | BLEU-4 | ROUGE-L | METEOR |
|---|---|---|---|---|---|---|
| Du et al. [3] | 2017 | BiLSTM encoder–decoder with attention | full SQuAD | 12.3 | 39.8 | 16.6 |
| Zhao et al. [10] | 2018 | Gated self-attention seq2seq with maxout pointer, paragraph level | full SQuAD | 16.4 | 44.5 | 20.3 |
| UniLM [4] | 2019 | Pre-trained unified Transformer | full SQuAD | 22.1 | 51.1 | 25.1 |
| ProphetNet [5] | 2020 | Pre-trained Transformer with n-stream decoder | full SQuAD | 25.0 | 52.6 | 26.8 |
| lmqg BART-base [6] | 2022 | BART encoder–decoder | full SQuAD | 24.7 | 52.7 | 26.1 |
| lmqg T5-small [6] | 2022 | T5 encoder–decoder, 60M parameters | full SQuAD | 24.4 | 51.4 | 25.8 |

| System | Year | Architecture | Training data | BLEU-4 | ROUGE-L | METEOR |
|---|---|---|---|---|---|---|
| lmqg T5-base [6] | 2022 | T5 encoder–decoder, 220M parameters | full SQuAD | 26.1 | 53.3 | 27.0 |
| This project, t5-small fine-tuned on CPU | 2026 | T5 encoder–decoder, 60M parameters | 8,000 SQuAD examples, 1 epoch | 17.0 | 42.8 | 41.5 |

Table 4. Published results on SQuAD compared with the models of this project.

The published rows reproduce the figures reported in the respective papers and model cards, which use the test split of Du et al. [3] and the nlg-eval scorers; the row for this project uses 300 paragraphs of the official development set with sacrebleu and rouge-score. The comparison therefore indicates trends rather than exact values. Scoring the lmqg checkpoints under the protocol of this project (24.5 BLEU-4 and 49.7 ROUGE-L for T5-base, against 26.1 and 53.3 on the model card) quantifies the offset between the two protocols at approximately 2 to 4 points. METEOR values are not comparable across rows because the NLTK implementation credits stem and synonym matches. Under this reading, the model fine-tuned on a CPU in under one hour is clearly above the 2017 LSTM system [3] and comparable to the 2018 paragraph-level system [10] (above it on BLEU-4, below it on ROUGE-L), while the T5-base checkpoint used in the demonstration is at the level of the strongest published SQuAD question generators.

## 9.3 Comparison of approach characteristics

| Approach | Requires a given answer | Verifies answerability | Item formats | Input handled |
|---|---|---|---|---|
| Rule-based transformation [2] | No; rules select trivial spans | Statistical ranking only | Wh-questions | Clean sentences |
| Seq2seq on SQuAD [3] | Yes | No | Wh-questions | Clean paragraphs |
| lmqg T5 checkpoints [6] | Yes, or a separate extractor | No | Wh-questions | Clean paragraphs |
| Prompting a hosted large language model | No | No; may hallucinate | Any | Pasted text; material leaves the device |

| Approach | Requires a given answer | Verifies answerability | Item formats | Input handled |
|---|---|---|---|---|
| StudyQG (this project) | No; hybrid answer selection with document coverage | Yes; DistilBERT round trip with a reported score | Short answer, multiple choice, cloze, true/false | PDF or text; offline on a CPU |

Table 5. Characteristics of existing approaches and of the proposed system.

### 9.4 Model selection

The task is the conditional generation of a short question from a long passage with one marked span. Encoder-only models such as BERT and DistilBERT [9] produce one representation per input token and cannot generate text; they are therefore used in this project for verification, where the output is a span of the input, rather than for generation. Decoder-only models of the GPT family can generate text but process the input strictly from left to right, so that the passage and the question compete for a single context window and the task must be conveyed through prompt wording. Encoder–decoder models encode the passage bidirectionally once and generate the question with a separate decoder that attends to every passage token through cross-attention; this family holds the published state of the art for SQuAD question generation [4, 5, 6].

Among encoder–decoder models, T5 [8] was selected for four reasons. First, its text-to-text formulation makes the task prefix and the `<hl>` highlight token a natural input representation and allows a single model to perform both answer extraction and question generation. Second, the model is available at sizes suited to the available hardware: t5-small (60 million parameters) was fine-tuned on a CPU in 45 minutes, and t5-base (220 million parameters) remains fast enough for a live demonstration. Third, public SQuAD checkpoints in exactly this input format [6] permit a controlled comparison between the model fine-tuned in this project and a full-data baseline. Fourth, pre-training on the C4 corpus provides strong transfer: 8,000 examples sufficed to raise BLEU-4 from 2.7 to 17.0. The trade-off is that a large hosted language model would produce more varied questions; the project prioritises a model that can be trained, measured and executed offline on the learner's own material.

## 10. Code Link

The complete source code, sample data, evaluation results, screenshots and this report are available at https://github.com/asil-khalifa/dl-project (public repository, branch main). The README documents installation, the demonstration and the reproduction of all

experiments. The fine-tuned checkpoint is not stored in the repository; it is reproduced by scripts/train.py in approximately 45 minutes on a CPU.

## 11. Novel Contribution

Published question-generation models address the mapping from a passage and an answer to a question. The contribution of this project is the complete system that makes such a model usable on real study material, together with a generator fine-tuned by the team.

1. Round-trip verified generation. Every generated question is answered by an independent extractive question-answering model, and only questions whose recovered answer matches the intended answer (token F1 of at least 0.5) are retained. The round-trip consistency criterion of Alberti et al. [7], originally proposed for filtering synthetic training corpora, is thereby applied as an interactive quality gate, and each item carries a quality score visible to the user.
2. Coverage-aware hybrid answer selection. Candidates are obtained from two sources (spaCy entities and noun chunks with a document-frequency salience score, and the answer-extraction mode of T5) and are selected round-robin across passages, so that a ten-question quiz spans the whole document rather than its first paragraph.
3. Four item formats from one generator. Distractors are drawn from spans of the same type within the same document (a question on Chandrayaan-3 is offered other ISRO missions rather than unrelated terms), with WordNet and numeric fallbacks; cloze and true/false items reuse the verified source sentence.
4. A generator fine-tuned by the team and a controlled comparison. A t5-small model was fine-tuned on 8,000 SQuAD examples in 45 minutes on a CPU and evaluated against an untrained baseline and the full-data lmqg checkpoints with BLEU-4, ROUGE-L and METEOR, separating the contribution of fine-tuning from that of model size and training data.
5. End-to-end PDF handling with traceable output. Every item carries its source sentence and a Bloom taxonomy level, and the entire pipeline executes on a laptop without a GPU.

## 12. Individual Contribution

| Member | Contribution |
|---|---|
| Asil Khalifa (106123015) | T5 question generator: highlight-format data preparation, fine-tuning of t5-small (scripts/train.py), beam-search generation (question_generator.py) and metric evaluation (scripts/evaluate.py) |
| L Hitesh Reddy (106123073) | DistilBERT round-trip verification: extractive question answering over the generated questions (quality_filter.py), selection of the acceptance threshold and integration into the pipeline |
| P Kamal Nag (106123093) | spaCy candidate answer extraction (answer_extraction.py) and answer comparison: SQuAD-style normalisation and the token-F1 score used to accept or reject each question |

Ingestion, item formatting, the web interface, the experiments and this report were shared work; all members reviewed the complete code base.

## 13. Learning Outcome

1. Fine-tuning a text-to-text Transformer for a new task: data formatting with task prefixes and highlight tokens, tokenisation, teacher forcing, cross-entropy loss, AdamW with warm-up, and checkpointing.
2. The distinction between training-time and inference-time behaviour of a sequence-to-sequence model, and the effect of decoding choices (beam size, repetition constraints) on the output.
3. Evaluation of generated text with BLEU-4, ROUGE-L and METEOR, the properties these metrics reward, and the need for a reference-free check such as round-trip question answering.
4. Use of pre-trained models from the Hugging Face Hub (lmqg T5, DistilBERT-SQuAD) as system components, and the trade-off between model size, training data and CPU latency.
5. The continued role of classical NLP tooling (spaCy named-entity recognition, noun chunks, WordNet) for answer selection and distractor generation around a neural model.

6. Engineering of a complete system: PDF ingestion, passage segmentation, batched CPU inference, a web interface, reproducible scripts and unit tests.

## 14. AI Tool Usage

| Tool | Purpose |
|---|---|
| Claude Code (Anthropic), model Claude Fable 5.1 | Scaffolding of the repository and drafting of the pipeline code, training and evaluation scripts, web interface, README, this report and the team's study notes, under the direction of the team. The team reviewed, executed and debugged the code and verified every reported result. The prompts issued are listed in Section 15. |
| Pre-trained models from the Hugging Face Hub | google-t5/t5-small [8], the lmqg T5 question-generation checkpoints [6] and DistilBERT-SQuAD [9], used as components of the system. |
| spaCy and NLTK WordNet | Rule-based NLP components; not generative AI. |

Table 6. AI tools used and their purpose.

No AI tool was used to produce or alter results. All reported metrics originate from executions of scripts/evaluate.py on the SQuAD validation subset, and all screenshots were captured from the running application.

## 15. Prompts Issued to the AI Agent

The team directed the AI coding agent (Claude Code) through a small number of deliberate prompts rather than many ad hoc requests. Each prompt records a decision taken by the team: what to build, which design to follow, what to measure, and what to change after inspecting the output. The agent produced code and drafts in response; the team reviewed, executed and verified everything. The prompts are reproduced below in consolidated form.

**Prompt 1. Framing the problem and the constraints**

We are Team 3 in CSPE72 Deep Learning Techniques; our assigned problem is Automatic Question Generation from Study Material. Before writing any code, read the assignment document in this folder and restate the deliverables and deadlines. We want a system that

a student can actually use: it must accept a lecture PDF or pasted notes and return examination-style questions with answers. Two constraints are fixed. First, the core must be a deep learning model that we fine-tune ourselves, so that we can explain and defend every training decision; a public checkpoint may be used only as a comparison point and as the demonstration default. Second, our only hardware is a 16-core laptop without a GPU, so any training run must finish within about one hour, and every number we report must be measured on this machine, never estimated or quoted from elsewhere.

**Prompt 2. Choosing the architecture**

We have decided on an answer-first pipeline rather than free generation: select what to ask about, then generate the question for that span. Implement it with an encoder–decoder model in the text-to-text style: T5 with the answer wrapped in highlight tokens and a task prefix, fine-tuned on SQuAD v1.1 in reverse (context and answer in, question out). Use t5-small for our own fine-tuning and keep the data preparation, training loop and tokenisation explicit in plain PyTorch rather than hidden behind a trainer class, because we will be asked to explain teacher forcing, the loss and the learning-rate schedule. Load the public lmqg T5 checkpoints through the same code so that either model can be selected at run time. Candidate answers should come from two sources, linguistic (spaCy entities and noun chunks, scored by type, syntactic role and document frequency) and neural (the answer-extraction mode of T5), and should be selected round-robin across passages so that the questions cover the whole document.

**Prompt 3. A quality gate instead of trusting the generator**

Fluent is not the same as correct. Add a verification stage: an independent extractive question-answering model (DistilBERT fine-tuned on SQuAD) answers every generated question from its passage, and the question is kept only if the SQuAD token-F1 between the recovered span and the intended answer is at least 0.5. Expose that score to the user as a quality column, rank by it, and discard any question that contains its own answer. From the verified items build four formats: short answer, multiple choice, fill in the blank and true/false. Distractors must be mined from the same document first (spans of the same entity type), with WordNet siblings and perturbed numbers or dates as fallbacks, so that a question about one ISRO mission is offered other ISRO missions rather than unrelated words.

**Prompt 4. Designing the evaluation**

Design the evaluation so that it isolates what each factor contributes. Evaluate four generators under one protocol on 300 held-out SQuAD paragraphs with beam size 4: t5-small without fine-tuning, our fine-tuned t5-small, and the lmqg t5-small and t5-base checkpoints. Report BLEU-4, ROUGE-L and METEOR, record seconds per question on our CPU, log the training and validation loss so that we can plot the curve, and save example predictions beside their references. When comparing with published results, state the

scorer and split differences explicitly and quantify the offset by scoring a public checkpoint under our own protocol; never present numbers as comparable when they are not.

**Prompt 5. Correcting the system after inspecting the output**

We ran the pipeline on our own sample texts and found specific defects; fix each one and show us the effect. Ordinal entities such as "first" become meaningless answers: exclude them. Neural answer spans sometimes carry trailing punctuation: strip it. Multiple-choice options for a place or a person were drawn from unrelated noun phrases: restrict fallbacks to proper-noun spans of similar shape, and generate date distractors by shifting the year or the month. The small model occasionally leaks the answer into the question: drop those questions. Separately, training and inference were an order of magnitude slower than expected; measure the effect of the PyTorch thread count on this machine and set the default accordingly.

**Prompt 6. Demonstration and submission**

Package the system for the demonstration and the written submission. Build a Gradio interface with file upload or pasted text, a model selector, controls for the number of questions and the item types, a results table with the quality score and source sentence, a quiz view and JSON/CSV export; capture screenshots from the running application. Write the report in a formal academic register following the assignment's section list exactly, with numbered sections, captioned figures and tables, numbered references and a comparison with published question-generation systems, and add the institute cover page from our earlier course report. Record which parts were human decisions and which were AI-assisted implementation, both in the report and in a study cheatsheet that explains the concepts, the code and likely viva questions in plain language.

Each prompt fixed a decision before implementation began; the role of the agent was to realise and document those decisions. Inspecting real output between prompts, as in Prompt 5, produced the largest gains in question quality.

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